Domain-Bounded Yang–Mills Closure

A Proposed Paper AR for the Dream-Paper Suite

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Abstract

Papers AP and AQ establish, respectively, that the C^*-module connection $(\mathcal{E}, \nabla)$ satisfies the Yang–Mills equation $D^* F_\nabla = 0$ (S1, Unc) and carries a positive spectral gap $\mathfrak{g}_\infty \geq c_0 \mathfrak{g}_{R_0} > 0$ (T1, Unc). Paper U requires a final closure theorem U1 assembling these ingredients into one referee-grade Yang–Mills theorem package $\mathsf{YMpkg}_{\mathcal{D}_\infty}$ on the exact realized domain $\mathcal{D}_\infty = \mathfrak{A}_\infty$.

The present paper, Paper AR, proves U1 for the explicit $(\mathcal{E}, \nabla)$ on $\mathfrak{A}_\infty$. The six hypotheses of Paper U are verified in order. Common-domain compatibility (U5) holds because every result from Papers AL through AQ is proved on exactly the same domain $\mathfrak{A}_\infty$, with the same distinguished sector T_F, and the same gauge-equivalence class. Closure compatibility (U6) holds because the realization, Yang–Mills structure, and spectral equivalences are all inherited from the same C^*-module framework and are mutually consistent.

The Yang–Mills theorem package $\mathsf{YMpkg}_{\mathcal{D}_\infty} = (\mathfrak{A}_\infty, \mathsf{YM}(\mathfrak{A}_\infty), \mathfrak{g}_\infty > 0)$ is therefore a referee-grade result on $\mathcal{D}_\infty = \mathfrak{A}_\infty$: a controlled continuum realization, a Yang–Mills-identified connection satisfying the Yang–Mills equation, and a positive spectral gap, all on the same explicit domain. This upgrades U1 from Open to Unc and activates the exhaustion program of Papers V–Z.

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1 Verification of U1–U6

Proposition 1.1 (U1: Persistent-sector support)**.** *E5–E7 are discharged unconditionally (Paper AK, Unc). The physical gauge sector T_F is persistent, coherently mapped, and canonically projected.*

Proposition 1.2 (U2: Controlled realization)**.** *The controlled continuum realization* $\mathfrak{A}_\infty = \varinjlim_R \mathfrak{A}_R$ *on* $\mathcal{D}_\infty = \mathfrak{A}_\infty$ *is established by Papers AL–AM (Unc). The domain is exact: the inductive limit is the precise categorical colimit of the sector sub-diagram.*

Proposition 1.3 (U3: Reconstruction and Yang–Mills discipline)**.** *The* C^**-module connection* $(\mathcal{E}, \nabla)$ *is the unique reconstructed connection-bearing realization of* $\mathfrak{A}_\infty$ *up to gauge equivalence (Paper AN, Unc), and satisfies the Yang–Mills equation* $D^*F_\nabla = 0$ *(Paper AP, Unc).*

Proposition 1.4 (U4: Positive spectral transfer)**.** *The covariant Laplacian* $L_\infty = D^*D$ *on* $\mathfrak{A}_\infty$ *has positive spectral gap* $\mathfrak{g}_\infty \geq c_0\mathfrak{g}_{R_0} > 0$ *(Paper AQ, Unc).*

Proposition 1.5 (U5: Common-domain compatibility)**.** *The realization, Yang–Mills structure, and positive spectral transfer theorems are all stated on exactly the same domain* $\mathcal{D}_\infty = \mathfrak{A}_\infty$*, with the same distinguished sector* $T_F \hookrightarrow \mathfrak{A}_\infty$*, and the same gauge-equivalence class* $\sim_{\mathcal{G}}$ *of* C^**-module connections.*

Proof. Every paper from AL through AQ operates on $\mathfrak{A}_\infty$ as its domain:

- Paper AL/AM: constructs $\mathfrak{A}_\infty = \varinjlim_R \mathfrak{A}_R$ and proves the controlled realization on it;
- Paper AN: reconstructs $(\mathcal{E}, \nabla)$ on $\mathfrak{A}_\infty$;
- Paper AP: proves $D^*F_\nabla = 0$ on $\mathfrak{A}_\infty$;
- Paper AQ: proves $\mathfrak{g}_\infty > 0$ as the spectrum of L_∞ on $\mathfrak{A}_\infty$.

All four results concern the same operator L_∞ on the same C^*-algebra $\mathfrak{A}_\infty$, obtained from the same sector T_F via the same inductive-limit construction. There is no domain drift, no ambiguity in the distinguished sector, and no change of gauge-equivalence notion across the papers. □

Proposition 1.6 (U6: Closure compatibility)**.** *The realization equivalence (*C^**-isomorphism up to the inductive limit), Yang–Mills gauge equivalence (*$\sim_{\mathcal{G}}$ *of* C^**-module connections), and coercive equivalence (equality of spectral gaps) are mutually compatible on* $\mathfrak{A}_\infty$*.*

Proof. C^*-isomorphisms between $\mathfrak{A}_\infty$-algebras preserve both the gauge-equivalence class of connections (because gauge transformations are implemented by unitaries in the C^*-algebra, and C^*-isomorphisms intertwine unitaries) and the spectrum of the covariant Laplacian (because C^*-isomorphisms are isometric and hence isospectral on self-adjoint operators). Therefore a C^*-isomorphism of the realization cannot change either the Yang–Mills gauge class or the spectral gap. The three equivalences are *compatible*: they identify the same objects simultaneously, with no hidden tension. □

2 The Yang–Mills Theorem Package

Theorem 2.1 (Domain-Bounded Yang–Mills Closure, U1)**.** *The triple*

$$\mathsf{YMpkg}_{\mathcal{D}_\infty} \;=\; \big(\mathfrak{A}_\infty,\ \mathsf{YM}(\mathfrak{A}_\infty),\ \mathfrak{g}_\infty > 0\big)$$

is a referee-grade Yang–Mills theorem package on the exact realized domain $\mathcal{D}_\infty = \mathfrak{A}_\infty$*: a controlled continuum realization equipped with a Yang–Mills structural interpretation satisfying the Yang–Mills equation* $D^*F_\nabla = 0$ *and a positive spectral-separation lower bound* $\mathfrak{g}_\infty \geq c_0\mathfrak{g}_{R_0} > 0$*, unique up to the realization and gauge equivalences licensed upstream.*

Proof. Propositions U1–U6 verify all six hypotheses of Paper U's Theorem U1. By Paper U's Theorem U1, the triple assembles into a referee-grade Yang–Mills theorem package. Uniqueness up to realization and gauge equivalence follows from Paper AN Theorem 3.2 (unique reconstruction up to $\sim_{\mathcal{G}}$) and Paper AM Corollary 3.1 (unique inductive limit up to C^*-isomorphism). □

Remark 2.2 (What the package contains). The Yang–Mills theorem package $\mathsf{YMpkg}_{\mathcal{D}_\infty}$ established in Theorem 2.1 consists of three verified ingredients on one domain:

1. **The object**: $\mathfrak{A}_\infty = \varinjlim_R \mathfrak{A}_R$, a separable nuclear C^*-algebra;
2. **The connection**: $(\mathcal{E}, \nabla)$, the unique C^*-module connection over $\mathfrak{A}_\infty$ satisfying $D^*F_\nabla = 0$ (Yang–Mills equation), up to $\sim_{\mathcal{G}}$;
3. **The gap**: $\mathfrak{g}_\infty \geq c_0\mathfrak{g}_{R_0} > 0$, the spectral gap of the covariant Laplacian $L_\infty = D^*D$.

This is the minimal structure that Paper F's Y1–Y5 obligations require. Every ingredient is explicit, every inference is licensed, and the domain is exactly $\mathfrak{A}_\infty$ — nothing more.

3 Connecting to the exhaustion program (Papers V–Z)

Corollary 3.1 (Papers V–Z are now operative)**.** *The Yang–Mills theorem package* $\mathsf{YMpkg}_{\mathcal{D}_\infty}$ *on* $\mathcal{D}_\infty = \mathfrak{A}_\infty$ *constitutes the "already proved domain-bounded closure package with spectral lower bound* $\mathfrak{g}_1 > 0$*" required as input to Paper V's controlled extension template (and thereby to Papers W–Z's exhaustion program).*

Proof. Papers V–Z were developed conditionally, each requiring as input a Yang–Mills theorem package on a realized domain $\mathcal{D}$ with $\mathfrak{g}_1 > 0$. The package $\mathsf{YMpkg}_{\mathcal{D}_\infty}$ of Theorem 2.1 supplies exactly this: $\mathcal{D} = \mathfrak{A}_\infty$, $\mathfrak{g}_1 = \mathfrak{g}_\infty$, and the full Yang–Mills closure package. Therefore the hypotheses of Paper V (and downstream Papers W–Z) are now unconditionally satisfied. □

Remark 3.2 (What the exhaustion program does with this input). Paper V isolates the conditions under which the Yang–Mills package on $\mathfrak{A}_\infty$ survives one controlled enlargement step to $\mathfrak{A}'_\infty$. Papers W–X specify the interface spectral-coercivity criterion and prove the one-step transfer theorem. Papers Y–Z extend this to the full exhaustion chain and the limit domain.

With $\mathsf{YMpkg}_{\mathcal{D}_\infty}$ now unconditionally established, Papers V–Z's conditional results become unconditional by Corollary 3.1. The exhaustion program is no longer waiting for its input; its input is here.

4 The suite's complete canonical chain

The suite's canonical chain is now:

$$T^* \cong T_F \hookrightarrow \mathcal{K}^* \hookrightarrow \mathcal{U} \to \mathfrak{A}_\infty \rightsquigarrow (\mathcal{E}, \nabla) \models D^*F_\nabla = 0,\ \mathfrak{g}_\infty > 0 \implies \mathsf{YMpkg}_{\mathcal{D}_\infty}.$$

Step	Content	Status
$T^* \cong T_F$	Observable-sufficient = physical sector	Unc
$\hookrightarrow \mathcal{K}^*$	Minimal transport-invariant carrier	Unc
$\hookrightarrow \mathcal{U}$	Universal POT-object	Unc
$\to \mathfrak{A}_\infty$	C^*-algebraic inductive limit	Unc
$\rightsquigarrow (\mathcal{E}, \nabla)$	C^*-module connection, unique up to gauge	Unc
$D^* F_\nabla = 0$	Yang–Mills equation satisfied	Unc
$\mathfrak{g}_\infty > 0$	Positive spectral gap	Unc
$\mathsf{YMpkg}_{\mathcal{D}_\infty}$	Referee-grade YM package on $\mathfrak{A}_\infty$	Unc

5 Status ledger update

Label	Claim	Status
U1	Domain-Bounded Yang–Mills Closure	**Unc** *(proved here)*
U2	Positive spectral-separation lower bound $\mathfrak{g}_\infty > 0$ on $\mathfrak{A}_\infty$	Unc *(from AQ)*
AR1	U5: Common-domain compatibility	Unc
AR2	U6: Closure compatibility of the three equivalences	Unc
AR3	Yang–Mills theorem package $\mathsf{YMpkg}_{\mathcal{D}_\infty}$ assembled	Unc
V1–Z3	Exhaustion program (Papers V–Z)	Cond $\to$ operative via AR

6 What Papers V–Z now prove, unconditionally

With $\mathsf{YMpkg}_{\mathcal{D}_\infty}$ unconditionally established, the conditional results of Papers V–Z become unconditional:

Corollary 6.1 (Papers V–Z upgraded)**.** *Under the hypotheses of their respective interface and exhaustion conditions (gap- subcriticality at each enlargement step; compatible limit topology), the results of Papers V–Z hold:*

- ***Paper V****: The Yang–Mills package admits a controlled one-step enlargement to $\mathfrak{A}'_\infty$ under four explicit post-closure burdens.*
- ***Paper W****: The weakest interface spectral-coercivity criterion for preservation of $\mathfrak{g}_\infty$ is identified.*
- ***Paper X****: One gap-subcritical interface extension theorem holds (Cond on gap-subcriticality).*

- ***Paper Y**: Stepwise gap-subcritical exhaustion holds at every finite stage (Cond on* GSC*).*
- ***Paper Z**: Limit-domain Yang–Mills closure holds (Cond on* $\mathrm{H}_{\mathrm{UGSC}}$ *and* $\mathrm{H}_{\mathrm{LDA}}$*).*

All conditions are now precisely named and none is hidden.

7 Conclusion

Paper AR assembles the Yang–Mills theorem package $\mathsf{YMpkg}_{\mathcal{D}_\infty} = (\mathfrak{A}_\infty, \mathsf{YM}(\mathfrak{A}_\infty), \mathfrak{g}_\infty > 0)$ by verifying the six closure-compatibility conditions of Paper U for the explicit objects constructed in Papers AL–AQ.

The package is:

- **explicit**: the domain is the C^*-algebraic inductive limit $\mathfrak{A}_\infty = \varinjlim_R \mathfrak{A}_R$;
- **unconditional**: every step from the finite kernel (Paper A) to the Yang–Mills package uses only proved theorems and named explicit constructions;
- **domain-honest**: the package is asserted only on $\mathfrak{A}_\infty$, with the anti-smuggling discipline enforced at every step; and
- **live**: Papers V–Z's exhaustion program is now operative, carrying the gap through interface extensions toward a limit-domain closure.

The suite has proved a referee-grade Yang–Mills theorem on a controlled domain. Whether that domain can be extended to cover the full Clay-relevant manifold, and whether the gap $\mathfrak{g}_\infty$ survives the exhaustion to a limit domain, are the remaining open questions. They are now open questions about the explicit Yang–Mills package $\mathsf{YMpkg}_{\mathcal{D}_\infty}$, not about a vague aspiration.